

# NICHOLAS A. HERNANDEZ

Atlanta, GA (can relocate) | (305) 484-9950 | nhern026@gmail.com | linkedin.com/in/nicholas-hernandez-bc | nhern026.github.io

## EDUCATION

### Georgia Institute of Technology

Atlanta, GA

*Master of Science in Computer Science - Computational Perception and Robotics*

*Fall 2026 - Expected May 2028*

- Current Coursework: Computer Vision, Deep Reinforcement Learning

### Boston College, Morrissey College of Arts and Sciences

Chestnut Hill, MA

*Bachelor of Science in Computer Science*

*Fall 2022 - Spring 2026*

- GPA: 3.91 | Summa Cum Laude | Phi Beta Kappa | Alpha Sigma Nu
- Relevant Coursework: Algorithms, Machine Learning, Robotics, NLP, Linear Algebra, Probability

## SKILLS

**Programming Languages:** Python, Java

**Machine Learning and Robotics:** PyTorch, Hugging Face, LeRobot, scikit-learn, NumPy, Pandas, OpenCV, MuJuCo

**Systems and Tools:** Linux/Unix, Git/GitHub, SLURM, Jupyter, XPolicyLab

## RESEARCH EXPERIENCE

### Georgia Institute of Technology, PAIR Lab

Atlanta, GA

*Research Engineer, advised by Dr. Animesh Garg*

*Fall 2026 - Present*

- Developing a bimanual robot-learning pipeline for two 6-DOF YAM arms using Meta Quest 3 VR teleoperation, with synchronized 100 Hz logging of joint states, commands, gripper signals, end-effector poses, and controller inputs.
- Building the demonstration-to-policy workflow for  $\pi_{0.5}$  fine-tuning with XPolicyLab and downstream real-robot evaluation on manipulation tasks in laboratory environments.

### Boston College

Chestnut Hill, MA

*Research Assistant, advised by Dr. Emily Prud'hommeaux*

*Summer 2025 - Spring 2026*

*Senior Honors Thesis: Cross-Linguistic Transfer Learning for Machine Translation*

*Fall 2025 - Spring 2026*

- Developed four transfer-learning pipelines for NLLB-200-distilled-600M, comparing single- and two-stage training and supplementary-language selection across 14 Formosan languages.
- Analyzed bidirectional BLEU results to identify the strongest training strategy; pooled high-resource Formosan data improved every evaluated target and increased FOR-to-EN BLEU by 50-79% for the four lowest-resource languages.

*NSF REU: Automatic Speech Recognition for Endangered Languages*

*Summer 2025*

- Developed parameterized Python and SLURM scripts to automate 80+ Wav2Vec2/XLSR-53 fine-tuning runs across donor-target pairs for 13 languages on a Linux compute cluster.
- Evaluated donor-selection strategies using cross-linguistic similarity metrics, reducing Word Error Rate by up to 68% and showing that unrelated-language transfer can outperform related-language transfer and additional in-language data.

### University of Houston

Houston, TX

*NSF REU Researcher, advised by Dr. Rakesh Verma*

*Summer 2024*

- Ran multilingual toxicity-detection experiments with XLM-RoBERTa across Romance languages; built an 85,000+ example dataset by sourcing, cleaning, and combining public corpora.

## PROJECTS

### SO-101 Robot Learning

*Python, LeRobot, Hugging Face, OpenCV*

*Fall 2026*

- Configured and calibrated paired SO-101 leader/follower arms with multi-camera LeRobot teleoperation; collected tabletop demonstrations, trained an ACT policy from scratch, and deployed it for closed-loop control on the physical robot.

### Robot Car Soccer System

*Python, OpenCV*

*Spring 2026*

- Designed and tested a real-time perception pipeline using HSV thresholding, Gaussian blur, and contour filtering to detect a ball while rejecting lighting artifacts and robot bodies.
- Automated goal and out-of-bounds event detection with multi-frame validation and point-in-polygon checks; integrated events with a Python game-state system and live Pygame scoreboard.

## PUBLICATION

- Prud'hommeaux, E., Scheppat, H., **Hernandez, N.**, & Le Ferrand, E. "A Family Affair: Counterintuitive Transfer Effects in ASR for Endangered Languages." *EMNLP 2026 Main Conference*.

## LEADERSHIP EXPERIENCE

### Cuban American Student Association

Chestnut Hill, MA

*President (senior year); Executive Board Member (all four years)*

*Fall 2022 - Spring 2026*

- Collaborated with peers, campus clubs, and the Office of Student Involvement to organize 10-15 cultural events per semester for 400+ members.